

Nathan Van Utrecht

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EDUCATION

University of California San Diego

M.S. Intelligent Systems, Robotics, and Control

San Diego, CA

Sep. 2025 – June 2027

Iowa State University

B.S. Mechanical Engineering; GPA: 3.98/4.00; Summa Cum Laude

Ames, IA

Aug. 2021 – May 2025

EXPERIENCE

Graduate Student Researcher

Xiaolong Wang's Lab, UC San Diego

Jan. 2026 – Present

San Diego, CA

- Co-designing dexterous hands and manipulation policies for Rubik's Cube solving on an Amazon-sponsored project; built MuJoCo/MJX environments and reward design for cube reorientation, flipping, and layer rotation
- Trained policies with JAX PPO on GPU, scaling a single cross-embodiment policy to 2,048 generated hand designs (4,096 parallel environments across 4 GPUs)
- Built the sim-to-real stack (domain randomization, AprilTag cube tracking, async policy server, Dynamixel control) and deployed the policy on a physical hand that solved a scrambled Rubik's Cube

Software Engineering Intern

Brain Corp

June 2026 – Sep. 2026

San Diego, CA

- Designed and shipped world-coordinate pallet-to-location association for a 4-camera warehouse robot, casting barcode reads as camera rays onto the rack face with WMS-aware arbitration; raised precision from 0.33 to 0.93 at 100% recall, now live at 10+ customer sites
- Built, as sole author, an evaluation platform (FastAPI, BigQuery, React) with a 25.7k-label golden set and commit-pinned pipeline runs; extended it into a harness for AI coding agents to hill-climb the pipeline, which found the shipped association change
- Built a barcode-constrained visual SLAM stitcher using sparse bundle adjustment over 426 poses, reaching about 1 mm accuracy on the rack and stitching a 139 m aisle 48× faster than the existing SIFT-based stitcher

Undergraduate Researcher (Honors Capstone)

Coordinated Systems Lab, Iowa State University

Nov. 2023 – May 2025

Ames, IA

- Benchmarked AIRL, GAIL, and Behavioral Cloning across 4 MuJoCo environments under shifted mass, friction, actuation, morphology, and goals; AIRL beat direct imitation by over 140% on transfer tasks
- Showed a transferred AIRL reward trains a new agent to 98% of oracle performance in a modified Hopper, and identified failure modes under out-of-distribution morphology and goal changes

REU Researcher

Translational Artificial Intelligence Center, Iowa State University

May 2024 – Aug. 2024

Ames, IA

- Benchmarked model-free SAC and PPO against model-based SHAC for sim-to-real robustness; SAC converged 5× faster and was more robust to system noise; reimplemented environments as differentiable simulators in PyTorch

PROJECTS

Visual-Inertial SLAM | EKF, Sensor Fusion, Stereo Vision

- Joint SE(3) EKF fusing IMU kinematics with stereo features on a Clearpath Jackal to bound dead-reckoning drift

LiDAR SLAM with Factor Graphs | GTSAM, ICP, Pose Graphs

- ICP scan matching and GTSAM pose-graph optimization; proximity loop closures cut graph error 65% vs 43% for fixed-interval

Data-Efficient Koopman Modeling with KANs | PyTorch, Koopman Operators, Soft Robotics

- KANs as learned Koopman observables for a soft robot arm; under half the training data of polynomial EDMD, about 2 ms per prediction

TECHNICAL SKILLS

Languages: Python, C/C++, TypeScript, MATLAB, SQL

Robot Learning: PyTorch, JAX, MuJoCo/MJX, Gymnasium, PPO, SAC, AIRL/GAIL, Sim-to-Real

Perception & Estimation: OpenCV, Camera Calibration, Bundle Adjustment, Visual SLAM, EKF, GTSAM, ICP

Infrastructure: GCP (Cloud Run, BigQuery), FastAPI, React, ROS, Docker, Git, Linux